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AN IN-SILICO INVESTIGATION OF RENAL ARTERY HEMODYNAMICS IN FENESTRATED ENDOVASCULAR ANEURYSM REPAIR USING THE LATTICE BOLTZMANN METHOD

Carter Groezinger, Arka Das, Leitao Chen

Department of Mechanical Engineering, Embry-Riddle Aeronautical University, Daytona Beach, FL, USA

ABSTRACT

Renal artery occlusion is a significant complication following fenestrated or branched endovascular aneurysm repair (F/B EVAR), often arising from adverse hemodynamic conditions induced by stent geometry and placement. Alterations in flow alignment, wall shear stress, and residence time within renal artery branches can promote thrombosis and compromise renal perfusion. Despite advances in endovascular stent design, the flow mechanisms governing post-operative occlusion remain incompletely understood.

This study presents a methodological assessment of a GPU-accelerated lattice Boltzmann method (LBM) framework for simulating renal artery hemodynamics in fenestrated or branched endovascular aneurysm repair (F/B EVAR). The goal is not to introduce a new physiological model, but to evaluate whether the proposed LBM approach can reproduce the principal branch-angle-dependent flow features reported in prior computational studies. A two-dimensional patient-generic model of the abdominal aorta and renal branch was implemented in FluidX3D, and steady-state simulations were carried out for multiple branch-angle configurations. The computed velocity fields captured the main qualitative hemodynamic behaviors observed in the reference study, including localized flow acceleration into the branch, near-ostial recirculation, and stronger flow separation at smaller branch angles. These results indicate that the proposed LBM framework can recover the dominant flow mechanisms governing disturbed renal branch hemodynamics in idealized F/B EVAR geometries. The findings support the suitability of GPU-accelerated LBM as an efficient alternative to conventional computational fluid dynamics for studying renal artery hemodynamics in F/B EVAR and provide a foundation for future three-dimensional, pulsatile, and patient-specific investigations.

Keywords: Lattice Boltzmann Method; Computational Hemodynamics; Endovascular Aneurysm Repair; Renal Artery Flow

1. INTRODUCTION

Endovascular aneurysm repair (EVAR) has become the preferred treatment for infrarenal abdominal aortic aneurysms due to lower perioperative morbidity and mortality than open repair [1]. However, standard EVAR is limited by the requirement for an adequate infrarenal neck, precluding its use in aneurysms extending to or involving the renal arteries. Fenestrated and branched endovascular aneurysm repair (F/B EVAR) was developed to address this limitation by enabling preservation of renal and visceral artery perfusion through fenestrations or directional branches incorporated into the main endograft [2–4]. These advanced techniques have demonstrated high technical success and acceptable mid- to long-term durability in appropriately selected patients [3,4].

Renal perfusion preservation remains a central determinant of procedural success in F/B EVAR. The kidneys receive approximately 20% of cardiac output, and renal blood flow is tightly regulated by autoregulatory mechanisms that maintain glomerular filtration across a range of perfusion pressures [5]. Even modest reductions in renal perfusion following aortic intervention are associated with deterioration in renal function, which is a strong independent predictor of long-term mortality [6]. Consequently, durable renal artery (RA) incorporation and branch stent patency are essential objectives in the endovascular treatment of complex abdominal and thoracoabdominal aortic aneurysms [5,6].

Despite technical advances, RA anatomy frequently limits the feasibility and durability of F/B EVAR. Comprehensive anatomic analyses have demonstrated that approximately one-fifth of patients present with absolute anatomic contraindications to renal artery incorporation, including small vessel diameter or early bifurcation, while an additional one-third exhibit challenging anatomy such as severe downward angulation, ostial stenosis, or prior renal stents [5]. Current device instructions for use typically recommend a minimum RA diameter of 4 mm and a length of at least 13 mm from the ostium to the first bifurcation to allow secure deployment and sealing of a bridging stent [5].

Failure to meet these criteria has been associated with increased procedural complexity and compromised branch patency.

Clinical outcome studies have confirmed that small renal artery targets are particularly vulnerable following F/B EVAR. Incorporation of RAs measuring less than 4 mm in diameter is associated with lower technical success, increased risk of arterial rupture, higher rates of branch instability, and significantly reduced primary and secondary patency at one year compared with larger renal arteries [8]. Although intentional coverage of small accessory renal arteries may be tolerated in selected patients, loss of a dominant renal artery or a substantial fraction of renal parenchyma is associated with ischemic injury and should be avoided whenever possible [5,7].

Beyond static anatomic suitability, branch stent geometry and deployment configuration play a critical role in determining post-operative hemodynamics. Fenestrated and branched stent grafts inherently alter flow patterns at the renal artery ostium due to protrusion of the bridging stent into the aortic lumen and redirection of inflow into the renal circulation [9,10]. In the idealized post-F/B EVAR configuration, flow near the renal ostium is influenced by the geometric configuration of the protruding branch stent within the abdominal aortic lumen. Computational investigations have demonstrated that unfavorable branch angles and excessive protrusion depths lead to disturbed flow, low wall shear stress, and flow recirculation at the renal artery origin, conditions known to promote platelet aggregation and thrombus formation [9,10]. These hemodynamic disturbances provide a mechanistic explanation for clinically observed branch occlusion despite technically successful deployment.

Traditional imaging modalities, such as computed tomography angiography, provide detailed anatomical assessment but are inherently limited in their ability to resolve local flow fields and shear stress distributions. Conventional computational fluid dynamics (CFD) approaches have yielded valuable insight into branch hemodynamics, but the cost of mesh generation and repeated high-resolution simulations can limit their efficiency in application-driven parametric studies. The lattice Boltzmann method (LBM) has been extensively validated in hemodynamic modeling and has consistently demonstrated good agreement with Navier-Stokes-based CFD solvers for arterial flow simulations. Previous investigations have shown that LBM is capable of reproducing physiologically relevant flow behavior in vascular domains, including velocity field development, near-wall flow structure, and pressure-driven transport in anatomically complex configurations [11-12]. Its mesoscopic formulation is particularly advantageous for capturing disturbed-flow features such as flow separation, recirculation zones, and stagnant regions, which are frequently encountered in branched arterial geometries and are directly relevant to thrombogenic risk assessment [12-15]. Owing to this combination of physical accuracy, geometric flexibility, and computational efficiency, LBM has emerged as a promising tool for the simulation of cardiovascular and endovascular flow environments.

Despite these advantages, the suitability of LBM for modeling renal branch flow after F/B EVAR must be established against a relevant benchmark before it can be adopted confidently for broader vascular studies. In particular, Wang et al. [9] reported angle-dependent renal artery flow features in an idealized post-F/B EVAR configuration, including branch-ostial recirculation, velocity redistribution, and changes in flow separation with branch angle. Reproducing those features provides a useful validation target for assessing whether an LBM-based solver can recover the same hemodynamic trends in a comparable geometry under comparable flow conditions.

Accordingly, the objective of the present study is to evaluate the ability of a GPU-accelerated LBM solver, FluidX3D, to reproduce the renal artery hemodynamic behavior reported by Wang et al. [9] in a patient-generic two-dimensional model. By focusing on branch-angle-dependent flow structure, including recirculation, separation, and velocity redistribution near the renal ostium, this work frames the problem as a methodological validation of LBM for post-F/B EVAR renal hemodynamics. Establishing such agreement is an essential step toward using GPU-accelerated LBM as an efficient alternative to conventional CFD for future parametric, three-dimensional, and patient-specific studies of complex endovascular repair.

2. METHODOLOGY

2.1 Governing Equations and Lattice Boltzmann Formulation

Blood flow was simulated using the lattice Boltzmann method (LBM), a mesoscopic numerical approach derived from kinetic theory that solves for the evolution of particle distribution functions (PDF) rather than directly discretizing the Navier-Stokes equations. In this work, the single-relaxation-time Bhatnagar Gross Krook (BGK) formulation was employed.

The discrete evolution equation for PDF or f_i is given by:

$$f_i(x + c_i, t + 1) = f_i(x, t) - \frac{1}{\tau}\left(f_i(x, t) - f_i^{eq}(x, t)\right) \tag{1}$$

where c_i are the discrete velocities, τ is the relaxation time, and f_i^{eq} is the equilibrium particle distribution function.

The relaxation time is related to the kinematic viscosity ν_{LB} in lattice unit by:

$$\nu_{LB} = c_s^2(\tau - 0.5) \tag{2}$$

where c_s is the speed of sound as $c_s = \frac{1}{\sqrt{3}}$ for D2Q9 lattice (Fig. 1). The D2Q9 lattice is a two-dimensional nine-velocity discrete velocity model, by which the original continuous particle velocity is presented by nine discrete particle velocities. Each discrete particle velocity, $\boldsymbol{\xi}_i$ (i alters from 0 to 8), is defined as

$$\boldsymbol{\xi}_i = \begin{cases} (0,0) & i=0 \\ \left(\cos\left[(i-1)\frac{\pi}{2}\right], \sin\left[(i-1)\frac{\pi}{2}\right]\right) & i=1-4 \\ \left(\cos\left[(i-5)\frac{\pi}{2}+\frac{\pi}{4}\right], \sin\left[(i-5)\frac{\pi}{2}+\frac{\pi}{4}\right]\right)\sqrt{2} & i=5-8 \end{cases} \tag{3}$$

The conversion from the discretized PDF to fluid density (ρ) is reduced to a summation, as given by Eq. 4 (a). Eq. 4 (b) recovers the momentum of the fluid from PDF and $\boldsymbol{\xi}_i$; the fluid velocity ($\boldsymbol{u}$) is calculated by dividing Eq. 4 (b) with ρ that is computed by Eq. 4(a).

$$\rho = \sum_{i=0}^{8} f_i \tag{4a}$$

$$\rho\boldsymbol{u} = \sum_{i=0}^{8} \boldsymbol{\xi}_i f_i \tag{4b}$$

Then the pressure field of the flow can be easily computed from the density field such as:

$$P = c_s^2\rho \tag{5}$$

which builds a linear relationship between pressure and density as it assumes in the kinetic theory that the pressure changes in the fluid flow are triggered by density changes. Therefore, Eq. (5) is known as the lattice equation of state.

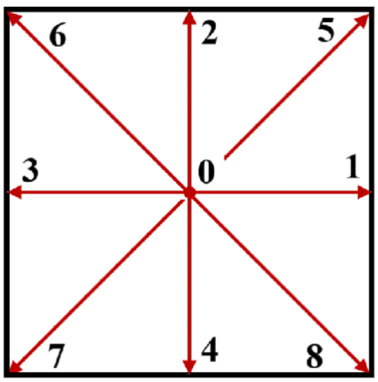


FIGURE 1: SCHEMATIC OF LBM LATTICE AND DISCRETE VELOCITY DIRECTIONS, D2Q9 LATTICE

Although Wang et al. [9] employed a three-dimensional model, the present study adopts a two-dimensional patient-generic geometry as a simplified framework for preliminary methodological assessment of the LBM approach. The objective is not to reproduce the full three-dimensional hemodynamics of the reference configuration exactly, but rather to determine whether the solver can recover the dominant branch-angle-dependent flow features, including flow separation, recirculation, and velocity redistribution near the renal ostium. This simplification reduces computational complexity and allows the influence of branch geometry to be examined in a more controlled manner. Accordingly, the present results should be interpreted as an initial validation of the numerical framework, with future three-dimensional simulations required to capture the full physiological flow structure.

2.2 Numerical Implementation Using FluidX3D

All simulations were performed using FluidX3D, a highly optimized, open-source LBM solver [13]. FluidX3D is designed specifically for GPU acceleration, enabling efficient simulation of large three-dimensional domains with minimal computational overhead.

Unlike conventional CFD solvers that require complex mesh generation and matrix assembly, LBM uses a structured lattice and local update rules, resulting in near-linear scaling with problem size. Previous studies have shown that GPU-accelerated LBM implementations can achieve orders of magnitude speedups compared to CPU-based solvers while maintaining numerical accuracy [14].

This computational efficiency is critical for the present study, as the Reynolds numbers relevant to renal artery flow following F/B EVAR need an extremely fine mesh resolution, which results in long simulation times, and then motivates the transition to a GPU accelerated solver like FluidX3D [15].

2.3 Patient-Generic Vascular Model Construction

Patient-generic models are commonly employed in vascular flow studies when the primary objective is to isolate and quantify the influence of selected geometric parameters, such as branch tilt angle, on local hemodynamics [16]. Unlike patient-specific reconstructions, which introduce substantial anatomical variability, idealized or patient-generic configurations provide a controlled framework for systematic parametric analysis. This enables clearer interpretation of cause-and-effect relationships between geometry and flow behavior, while also facilitating direct comparison with prior theoretical and computational studies.

A patient-generic vascular geometry was constructed to replicate the theoretical model described by Wang et al. [9]. The geometry consists of a straight aortic segment with a single branch representing a renal artery stent.

The abdominal aorta was modeled as a channel-like, two-dimensional conduit with diameter D_1, while the branch stent was represented as a cylindrical conduit of diameter D_2, oriented at a prescribed tilt angle θ relative to the aortic axis, which is shown in Fig. 2

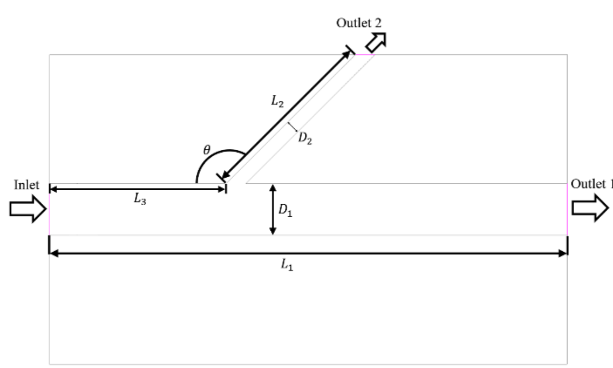


FIGURE 2: GEOMETRY SCHEMATIC SHOWING D_1, D_2, L_1, L_2, L_3, and θ

2.4 Boundary Conditions and Flow Driving Mechanisms

Unlike traditional CFD solvers, which impose boundary conditions directly on macroscopic variables such as pressure and velocity, the lattice Boltzmann method operates on PDFs.

Because the governing equation in LBM solves PDFs rather than the macroscopic variables directly, physical boundary conditions must be interpreted and implemented in terms of the PDFs on the boundary.

In this study, flow within the abdominal aorta was driven by pressure-type boundary conditions implemented through prescribed density values. In LBM, pressure and density are linked through the lattice equation of state, Eq. (5), so a prescribed density corresponds to a prescribed pressure. Constant but different density values were therefore applied at the inlet and at the two outlets (refer to Fig. 2), generating the pressure gradient required to drive the flow and allowing flow to partition naturally between the aortic outlet and the renal branch outlet.

All vessel walls were modeled as no-slip boundaries. Whereas traditional CFD enforces no-slip by directly imposing zero velocity at the wall, LBM imposes this condition through bounce-back treatment of the PDFs at solid boundaries. This produces an effective zero-velocity condition at the wall and is widely used in hemodynamic simulations to capture near-wall flow structure and wall shear behavior [11].

2.5 Simulation Parameters and Post-Processing

The simulation parameters were selected to match, as closely as possible, those reported in the reference study of Wang et al. [9], so that the present work could serve as a methodological comparison of the lattice Boltzmann framework rather than an independent redefinition of the hemodynamic problem. In particular, the geometric configuration, branch-angle cases, and flow conditions were prescribed consistently with the F/B EVAR model described in [9]. Fluid properties and nondimensional flow characteristics were chosen to reproduce the same hemodynamic regime, and the boundary driving conditions were defined to yield comparable flow partitioning and branch-ostial behavior. By maintaining consistency with the reference setup, differences in the computed flow field can be directly attributed to the difference in methodology rather than to changes in the physical parameter setup in the model. This parameter alignment is therefore essential to evaluating whether the present LBM implementation can reliably reproduce the key renal artery hemodynamic trends reported in [9].

Simulations were performed until steady-state conditions were achieved, as determined by convergence of velocity and flow rate metrics. Flow fields were exported in VTK format and analyzed using ParaView. Velocity magnitude contours and streamline visualizations were generated to identify regions of flow separation and recirculation.

3. RESULTS AND DISCUSSION

3.1 Flow Field Characteristics in the Patient-Generic Model

Steady-state simulations were performed at a Reynolds number of Re=7200, consistent with the inlet conditions reported by Wang et al. [9], where Re in the LBM simulations is defined as:

$$Re = \frac{u_{LB} L_{LB}}{\nu_{LB}} \qquad (6)$$

where u_{LB} is the characteristic velocity in lattice unit measured in the LBM simulation (we picked the value at the inlet); L_{LB} is the characteristic length in lattice unit (we picked D_1); ν_{LB} is computed by Eq. (2). In the LBM framework, the flow is solved in lattice units rather than directly in physical units such as velocity, pressure, and viscosity. As a result, these quantities are represented internally in a scaled numerical form and are not uniquely associated with a single physical case unless an additional dimensional mapping is introduced. The physically meaningful connection between the LBM simulation and the reference F/B EVAR problem is therefore established through dimensionless similarity parameters. Therefore, the Reynolds number is used as the primary matching criterion. By matching the Reynolds number measured in [9], the present simulations ensure dynamic consistency with the reference case even though the internal LBM variables are expressed in lattice units rather than explicit physical values.

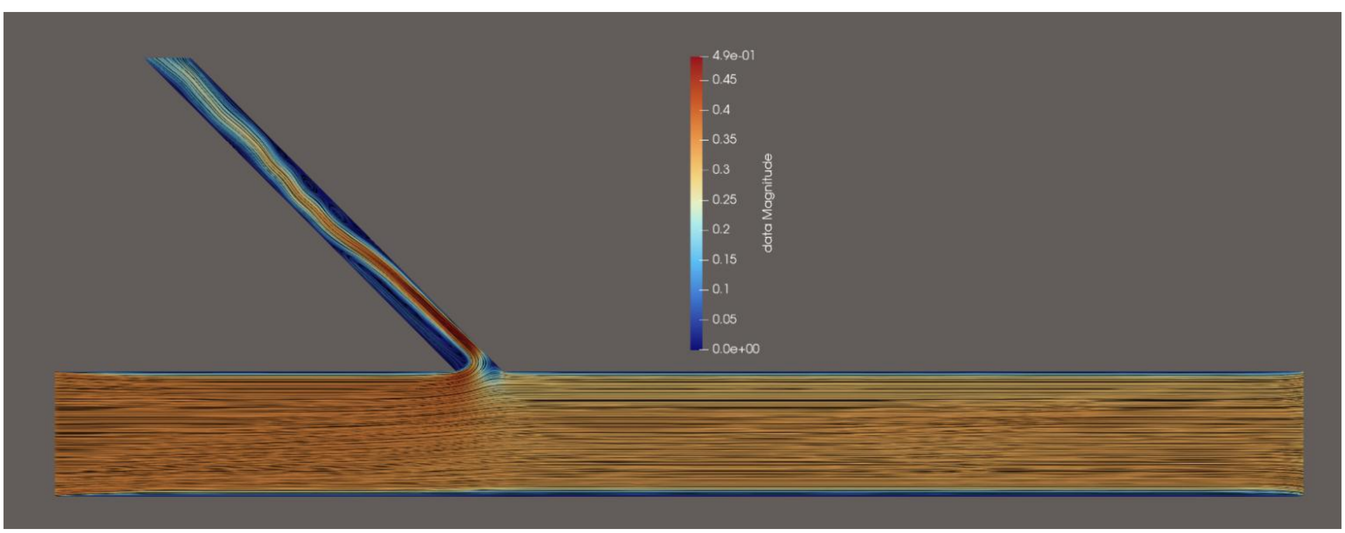


FIGURE 3: VELOCITY SREAMLINE PLOT FOR $\theta = 45°$

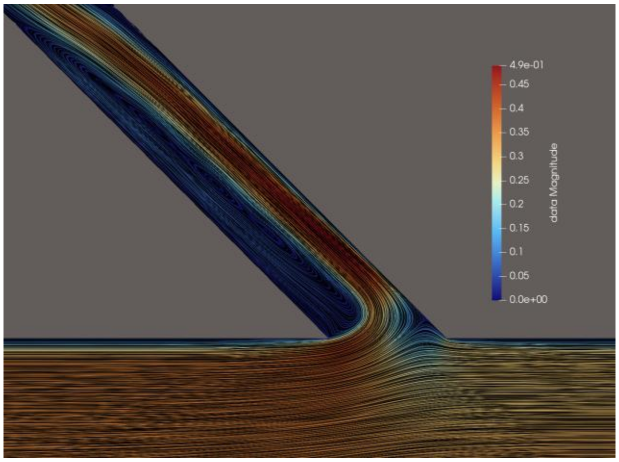


FIGURE 4: VELOCITY STREAMLINE PLOT OF THE BRANCH OSTIUM REGION FOR $\theta = 45°$, HIGHLIGHTING RECIRCULAITON

Flow was driven by prescribed density boundary conditions at the inlet and both outlets, with ρ_{in} = 3.25, ρ_{out1}=0.50, and ρ_{out2} = 0.85 (all values are lattice units), corresponding to lattice pressures of p_{in}=1.083, p_{out1}=0.167, and p_{out2}=0.283 via the D2Q9 equation of state $p = c_s^2\rho = \rho/3$, and a total aortic driving pressure differential of Δp=0.917 in lattice units. The branch-to-aorta diameter ratio was D_2/D_1 = 0.267, consistent with Wang et al. [9]. The final mesh resolution has 27,000 lattice nodes in L_1 and 2,700 lattice nodes in D_1, yielding approximately 72 million total lattice nodes.

All results correspond to fully converged steady-state solutions confirmed by monitoring velocity residuals and flow rate conservation across both outlets. Figure 3 illustrates the velocity magnitude field within the abdominal aorta-branch configuration for a branch angle of $\theta = 45°$. The highest velocities are observed in the core of the main aortic channel, with progressive decay toward the walls due to the no-slip condition. Near the renal branch junction, the flow is redirected into the angled branch, producing a localized acceleration within

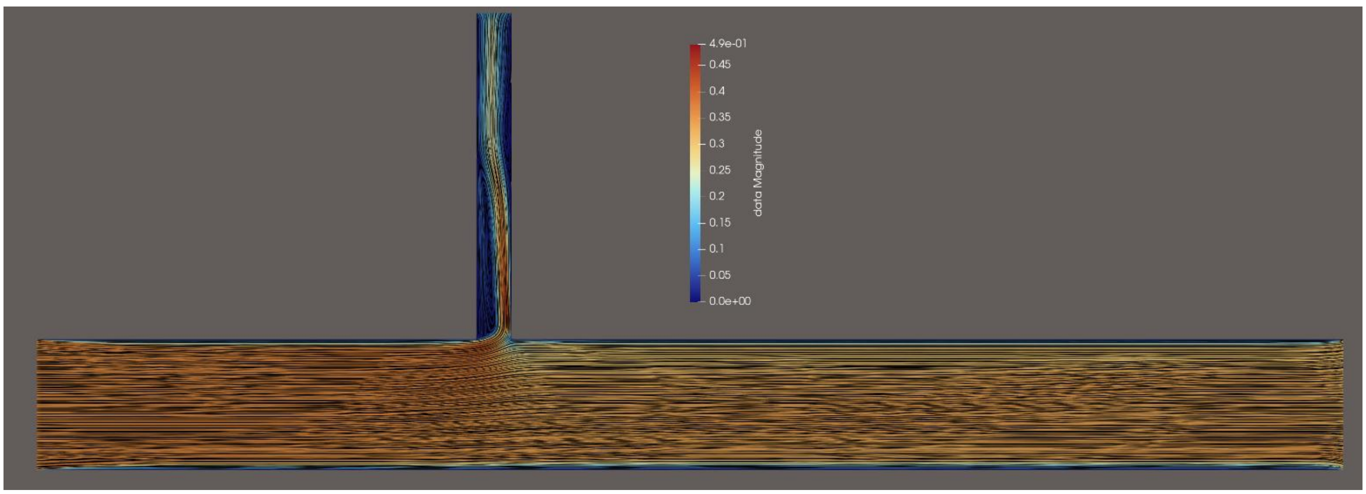


FIGURE 5: VELOCITY SREAMLINE PLOT FOR $\theta = 90°$

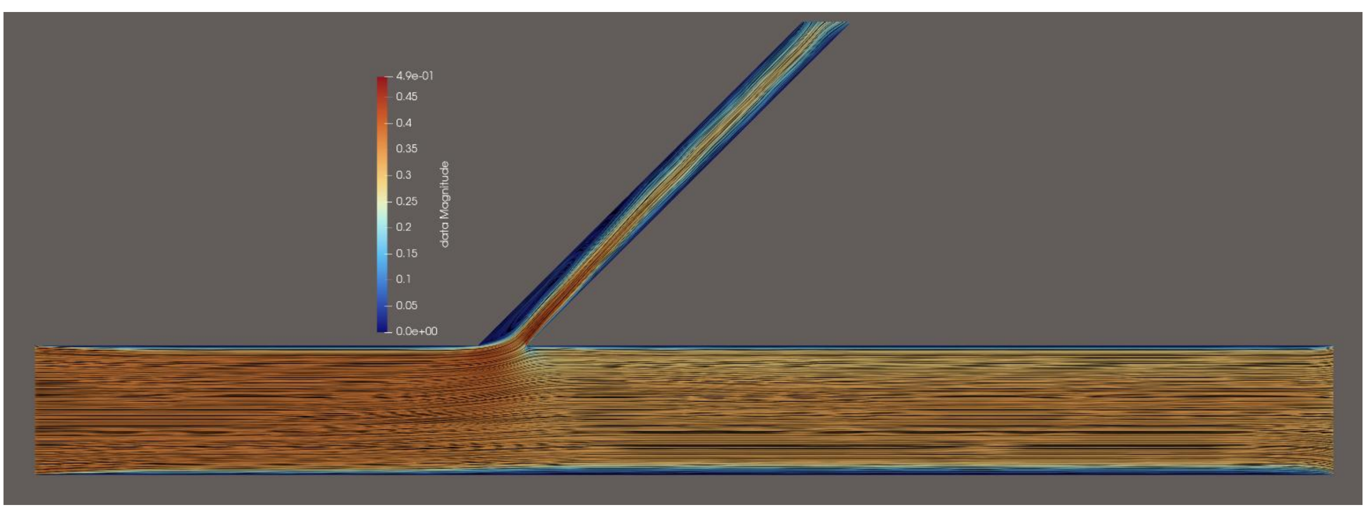


FIGURE 6: VELOCITY SREAMLINE PLOT FOR $\theta = 135°$

the branch lumen. However, this turning motion also generates flow separation at the branch ostium, where velocity vector and streamline patterns reveal a recirculation zone along the downstream wall of the branch entrance. This behavior is shown more clearly in the close-up view of the stent region in Fig. 4, which highlights the low-velocity recirculating flow and locally reversed flow direction relative to the main aortic stream. A separation bubble forms immediately downstream of the branch entry, indicating flow misalignment induced by the branch angle. These results show that the branch geometry produces disturbed near-ostial hemodynamics, and the predicted location and extent of the recirculation zone are consistent with those reported in the reference study. Details on the comparison with the reference [9] are discussed in Sec. 3.2.

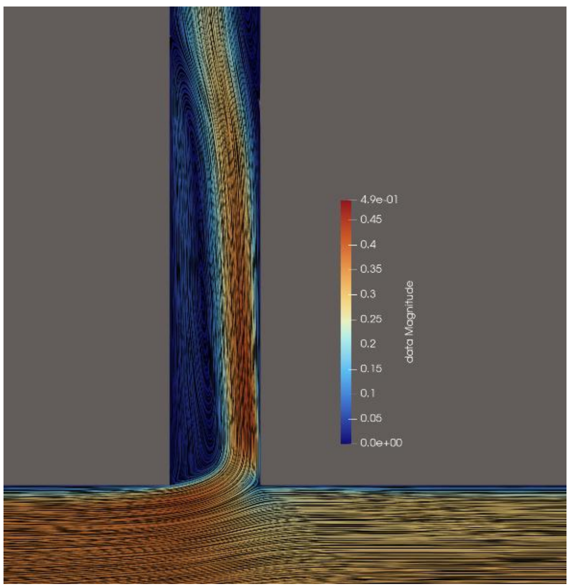


FIGURE 7: VELOCITY SREAMLINE PLOT OF THE BRANCH OSTIUM REGION FOR $\theta = 90°$

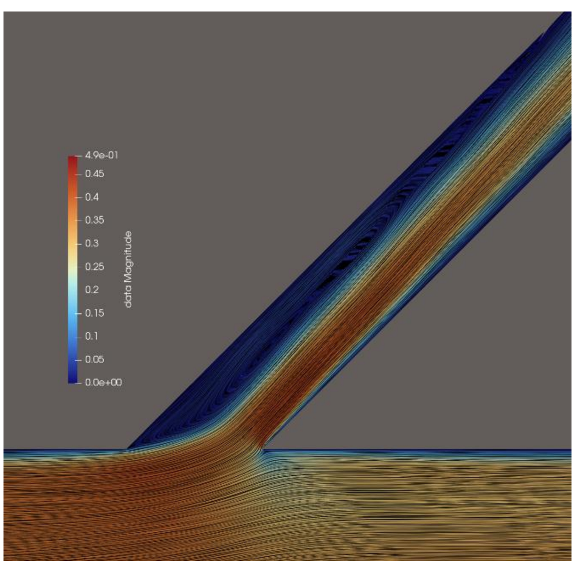


FIGURE 8: VELOCITY SREAMLINE PLOT OF THE BRANCH OSTIUM REGION FOR $\theta = 135°$

3.2 Effect of Branch Angle on Local Hemodynamics

Simulations were repeated for multiple branch angles to assess the sensitivity of local hemodynamics to stent orientation. The influence of branch angle on the local hemodynamics is illustrated in Figs. 3 to 8, which show the velocity field and near-ostial flow structure for the different branch-angle configurations considered in the present study. Across all cases, the bulk of the flow remains concentrated within the core of the main abdominal aortic passage, while the branch geometry locally redirects the incoming flow into the renal branch. However, the degree of flow disturbance near the branch ostium varies markedly with branch angle.

For the smaller-angle configurations shown in Figs. 3 and 4, the incoming abdominal aortic flow must undergo a sharper directional change to enter the branch, which promotes stronger flow misalignment at the ostium. As a result, a broader low-velocity region develops near the downstream wall of the branch entrance, accompanied by a more pronounced recirculation zone and a larger separation bubble. These features indicate that smaller branch angles create a less favorable local flow environment, with a greater portion of the near-ostial region occupied by slowly moving and recirculating fluid.

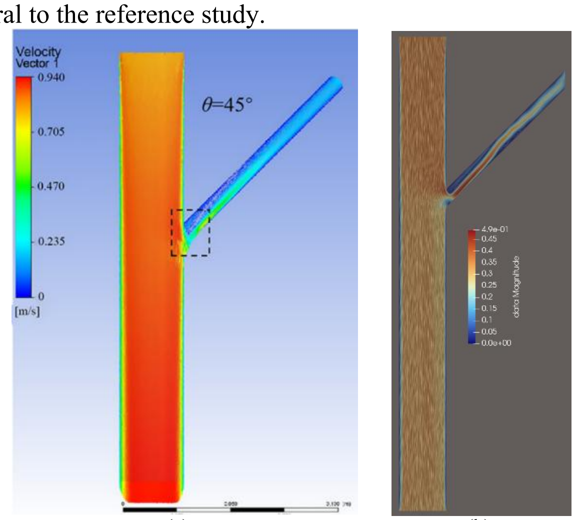
As the branch angle increases, as illustrated in Figs. 5 and 6, the turning of the incoming flow becomes less abrupt and the momentum transfer from the abdominal aortic stream into the branch becomes more direct. In these intermediate-angle cases, the recirculation zone remains present but becomes more confined to the immediate vicinity of the branch ostium. The velocity field within the branch also appears more uniformly distributed, suggesting improved flow attachment along the branch wall and reduced separation at the entrance.

For the larger-angle configurations shown in Figs. 7 and 8, the branch inflow exhibits the most streamlined behavior among the cases considered. The separated region near the ostium is reduced further, and the low-velocity zone adjacent to the downstream wall occupies a smaller spatial extent. These results indicate that larger branch angles are associated with diminished recirculation and improved branch inflow characteristics, as the flow is able to enter the branch with less severe turning-induced disturbance.

3.3 Comparison with Published Computational Results

Comparison of the present LBM simulations with the published computational results of Wang et al. [9] shows that the current model reproduces the principal flow features reported for the F/B EVAR renal branch configuration. In both the global velocity-magnitude plots and the close-up views of the branch ostium, the present results exhibit the same characteristic flow behavior observed in the reference study, including a high-velocity core within the abdominal aortic lumen, acceleration of the flow as it enters the renal branch, and the formation of a low-velocity recirculation region adjacent to the downstream wall of the branch entrance. The predicted location of the separation bubble near the ostium is also consistent with the published results, indicating that the present LBM implementation captures the dominant disturbed-flow structures associated with the angled branch geometry.

The comparison across the different branch-angle cases further supports this agreement. As shown in Figs. 9 to14, the present simulations recover the same qualitative angle-dependent trends reported by Wang et al. [9], namely that reduced branch angle is associated with more pronounced near-ostial flow disturbance, a larger separated region, and less favorable branch inflow characteristics. Conversely, as the branch angle increases, the present results show improved flow attachment and a reduction in the spatial extent of the recirculating zone. The ability of the LBM model to reproduce these systematic trends is important because it indicates that the solver captures not only the general appearance of the flow field, but also the geometric sensitivity of the hemodynamics that is central to the reference study.

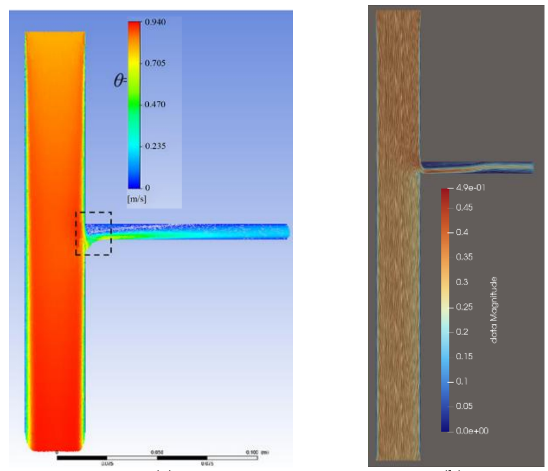


(a) (b)

FIGURE 9: COMPARISON TO REFERENCE VELOCITY MAGNITUDE PLOT AT $\theta = 45°$. (a) Wang et al. [9]; (b) Current study

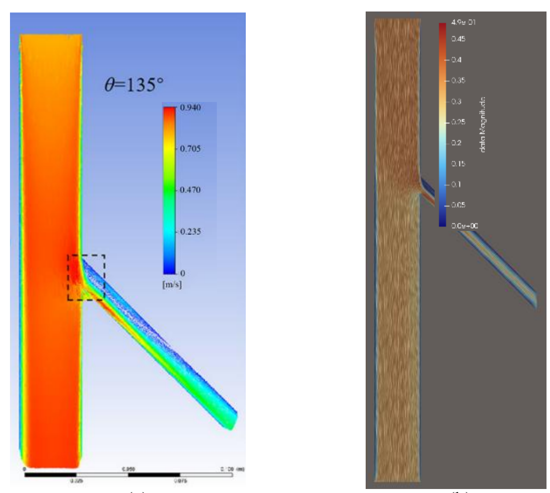
Some differences remain in the local distribution of velocity magnitude, particularly near the branch outlet and in portions of the downstream branch lumen. These differences are likely attributable to methodological differences between the present LBM formulation and the conventional Navier-Stokes-based approach used in the reference study, as well as to differences in how the boundary conditions are represented in the two numerical frameworks, which is also methodological. In particular, the present simulations impose pressure-type driving through lattice-density boundary conditions, whereas conventional CFD formulations typically prescribe macroscopic pressure or flow variables directly. Such differences can influence the detailed local velocity field even when the dominant global flow structures remain similar.

Quantitative comparison with [9] is limited by the fact that the published results are reported in physical quantities, such as mass flow rate, while the present LBM simulations are formulated entirely in lattice units. Because LBM does not directly produce dimensional physical outputs unless an additional scaling procedure is introduced, the mass flow rates obtained in the reference study cannot be compared directly to the raw simulation values reported here. For this reason, the present comparison is focused on qualitative agreement in the dominant hemodynamic features, including recirculation, flow separation, and velocity redistribution, rather than on direct numerical comparison of dimensional flow variables.



(a) (b)

FIGURE 10: COMPARISON TO REFERENCE VELOCITY MAGNITUDE PLOT AT $\theta = 90°$. (a) Wang et al. [9]; (b) Current study

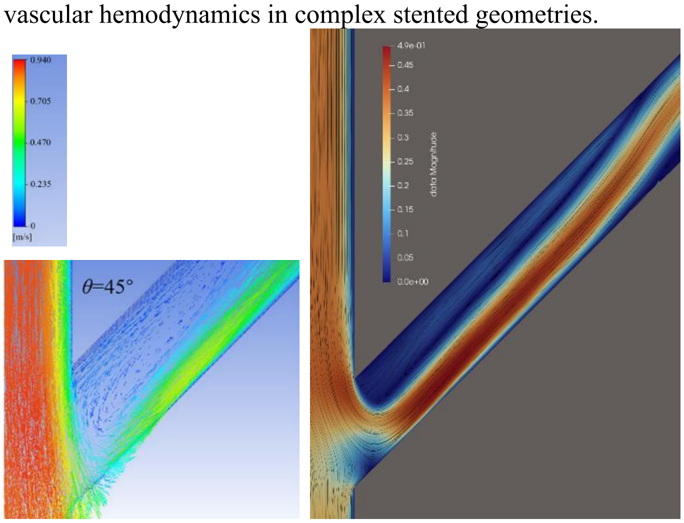


(a) (b)

FIGURE 11: COMPARISON TO REFERENCE VELOCITY MAGNITUDE PLOT AT $\theta = 135°$. (a) Wang et al. [9]; (b) Current study

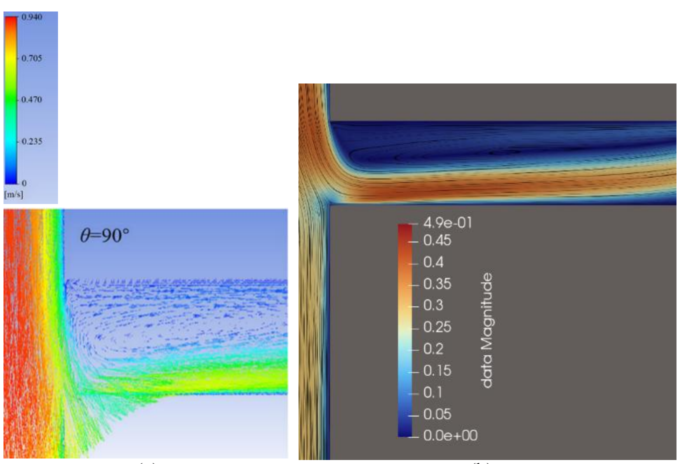
4. CONCLUSION

This study demonstrates that a GPU-accelerated LBM framework can reliably capture key hemodynamic features associated with renal artery branch stents following fenestrated endovascular aneurysm repair. Using a patient-generic model, the simulations reproduced flow separation, recirculation zones, and branch angle dependent trends that are consistent with previously published computational investigations [9]. The observed agreement supports the validity of LBM as a high-fidelity alternative to conventional Navier-Stokes solvers for vascular hemodynamics in complex stented geometries.



(a) (b)

FIGURE 12: COMPARISON TO REFERENCE VELOCITY MAGNITUDE PLOT OF THE BRANCH OSTIUM AT $\theta = 45°$. (a) Wang et al. [9]; (b) Current study

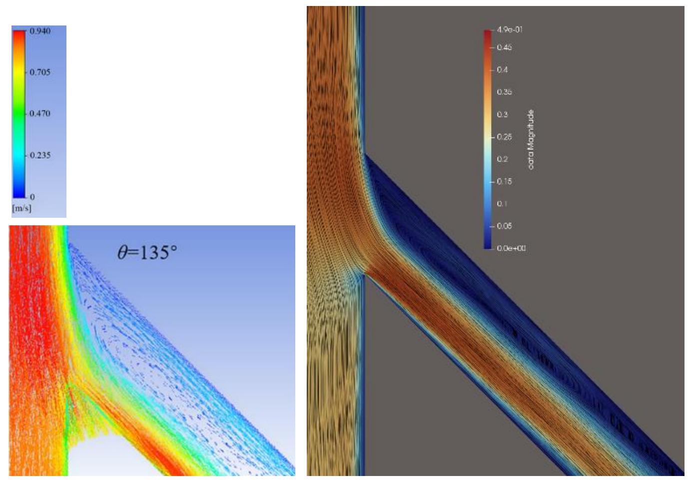


(a) (b)

FIGURE 13: COMPARISON TO REFERENCE VELOCITY MAGNITUDE PLOT OF THE BRANCH OSTIUM AT $\theta = 90°$. (a) Wang et al. [9]; (b) Current study

The present work highlights the potential of LBM-based modeling as a valuable tool for device evaluation and procedural planning in fenestrated and branched endovascular aneurysm repair. Future extensions to three-dimensional, pulsatile, and non-Newtonian flow simulations will further enhance the physiological relevance of this approach.

These in silico findings also highlight the physiological sensitivity of the renal ostial flow field to the implanted branch-stent configuration following F/B EVAR. In the patient-generic model, the reproduced separation and recirculation patterns illustrate how the protruding stent geometry within the abdominal aortic lumen can influence the manner in which abdominal aortic flow is conveyed to the renal branch, thereby affecting the local hemodynamic flow field that supports branch perfusion. Accordingly, the present work provides clinically meaningful support for continued computational study of post-implant renal branch hemodynamics and offers a practical framework for extending such analyses to more advanced vascular models.



(a) (b)

FIGURE 14: COMPARISON TO REFERENCE VELOCITY MAGNITUDE PLOT OF THE BRANCH OSTIUM AT $\theta = 135°$. (a) Wang et al. [9]; (b) Current study

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